Embedding-Based Web Text Triage for Ancient Chinese Paleography

1 Research Question

This project asks whether an embedding-based classifier can help triage short web snippets for building a comprehensive corpus of Ancient Chinese paleography. Relevant resources are spread across article sites, character-reference databases, and transcription repositories, so corpus construction requires deciding not only whether a snippet is relevant, but also what kind of material it is. We formulate this as short-text classification: given a snippet from a public web source, the system predicts whether it is scholarly discussion, primary transcription, dictionary/reference content, or noise. The goal is not to replace expert interpretation, but to provide a practical first-pass routing tool.

2 Dataset and Labels

Because large-scale web scraping can be ethically and technically ambiguous, this project uses 400 manually collected and annotated snippets instead of an automated crawler. Each row contains the text, label, source URL, and language. Sources include major paleography and character-reference websites [1–4]; the full source list is included in `source_used.md`.

The final label set contains four classes, shown in Table 1. We also considered adding a separate metadata or catalogue-records class, such as object numbers, collection identifiers, bibliographic rows, database fields, and other structured records. We did not include it in the final scheme because this kind of information is highly structured rather than ordinary raw text. It is usually better handled by field extraction, table parsing, or source-specific database processing than by semantic short-text classification.

Label	**Meaning**
`kpt`	Primary transcription material, such as oracle-bone divinations, bronze inscriptions, excavated manuscript fragments, or other source-text transcriptions.
`kde`	Dictionary or reference entries, including character forms, meanings, pronunciations, glosses, variants, and cross-references.
`ksd`	Scholarly discussion, interpretive prose, research summaries, or paleographic argumentation.
`noise`	Navigation text, dates, login prompts, advertisements, unrelated text, broken OCR, or other material not suitable for the corpus.

Table 1: Final label set.

The dataset is split into train, validation, and test sets with a 70/15/15 ratio: 280 training examples, 60 validation examples, and 60 test examples. Each class contains 100 examples in total and remains balanced across the splits.

3 Method

The project uses `intfloat/multilingual-e5-small` to encode each snippet as a normalized sentence embedding. This model fits the task because the data is mainly Chinese but may later expand to English or Japanese scholarly discussion, and because it is lightweight enough for local experiments and Hugging Face Space deployment.

On top of the embeddings, we compared three simple classifiers: an average embedding baseline, logistic regression, and linear SVM. The average baseline assigns each example to the nearest class centroid; logistic regression and linear SVM are strong linear models for high-dimensional text features. The final demo uses logistic regression because it tied with linear SVM on the test set while also providing class probabilities for the Gradio interface. We did not use a neural classifier because the dataset has only 400 examples, making overfitting a serious risk.

4 Evaluation

The final logistic regression model achieved 0.9333 accuracy and 0.9298 macro F1 on the held-out test set, with 4 errors out of 60 examples. Figure 1 summarizes the per-class scores, confusion matrix, and t-SNE projection. The model performs very well on `kde`, `kpt`, and `noise`; all test errors come from `ksd`.

Class	P	R	F1
kde	0.882	1.000	0.938
kpt	0.938	1.000	0.968
ksd	1.000	0.733	0.846
noise	0.938	1.000	0.968
Macro	0.939	0.933	0.930

Gold/Pred.	kde	kpt	ksd	noise
kde	15	0	0	0
kpt	0	15	0	0
ksd	2	1	11	1
noise	0	0	0	15

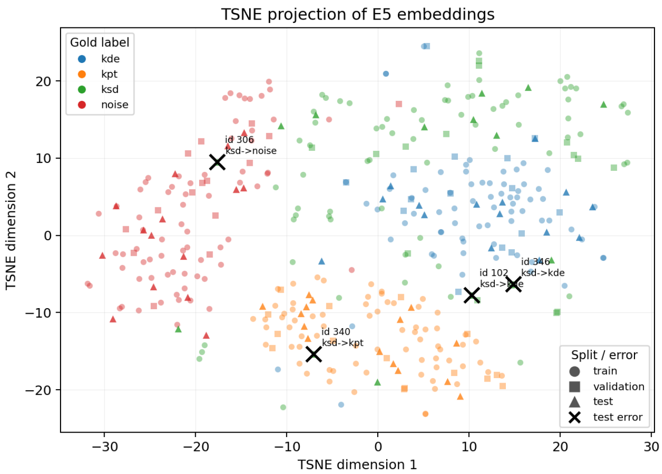

Figure 1: Test-set evaluation and t-SNE visualization. Black crosses mark the four logistic-regression test errors.

5 Error Analysis and Limitations

All 4 test errors are `ksd` examples predicted as another class: 2 as `kde`, 1 as `kpt`, and 1 as `noise`. Figure 1 helps explain this pattern: `noise`, `kpt`, and `kde` form relatively visible regions, while `ksd` is more scattered and often lies between neighboring categories. This is plausible because scholarly prose may contain character explanations, dictionary-like conclusions, quotations of primary material, or short contextual notes.

The main limitations are the small dataset size, the Chinese-centered training data, and fuzzy label boundaries. The model may not generalize well to English or Japanese scholarly discussion, and isolated snippets lose useful signals from page structure, titles, surrounding paragraphs, and source-site information.

6 Deployment and AI Tool Reflection

The project has been published as a Hugging Face Dataset, Model, and Space demo, with code and documentation stored in GitHub. AI assistance was used for scripting, evaluation, the Gradio demo, visualization, LaTeX formatting, and English report drafting. This greatly improved efficiency and let me focus more on annotation design and theoretical analysis, especially because English is not my first language. Discussing errors with AI was also useful: its explanation of `ksd` confusions matched both my own interpretation and the t-SNE plot. However, this also shows the need for caution. Plausible AI output must be checked against data, code, and visual evidence. Overall, AI can speed up research, but it requires strong theoretical judgment, basic practical skills, and attention to ethical issues such as data handling, interpretation, and responsibility.

- Hugging Face Dataset: `https://huggingface.co/datasets/Harry214/paleography-web-text-triage-dat`
- Hugging Face Model: `https://huggingface.co/Harry214/paleography-web-text-triage-logreg`
- Hugging Face Space: `https://huggingface.co/spaces/Harry214/paleography-web-text-triage-demo`
- GitHub repository: `https://github.com/hytung214-ai/paleography-web-text-triage`

References

- [1] *Lexicon of Pre-Qin Oracle, Bronze Inscriptions and Bamboo Scripts*.
- [2] Xiaoxuetang.
- [3] Center for Research on Chinese Excavated Classics and Paleography, Fudan University.
- [4] *Multi-function Chinese Character Database*.
- Full source list: `source_used.md`.